

FAIR ISLE, United Kingdom

WMO: 030080

Lat: 59.526N

Lon: 1.627W

Elev: 57

StdP: 100.64

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.4	0.7	-5.9	2.3	1.7	-4.8	2.5	2.1	21.4	5.0	20.1	4.9	8.9	330	0.886

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.9	15.6	14.4	14.7	13.6	14.2	13.2	14.7	15.3	14.0	14.4	13.4	13.9	5.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	10.2	14.9	13.7	9.8	14.3	13.1	9.5	13.7	41.2	15.4	39.4	14.5	38.0	14.0	17.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.5	14.8	DB	-1.6	17.4	1.5	1.5	-2.7	18.5	-3.6	19.4	-4.5	20.2	-5.6	21.3	(4)
(5)			WB	-3.4	15.5	1.6	1.2	-4.5	16.4	-5.5	17.1	-6.3	17.8	-7.5	18.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	5.3	4.8	5.2	6.2	7.9	10.1	11.9	12.5	11.5	9.6	7.4	6.1
	DBStd	3.21	1.81	2.09	2.03	1.77	1.65	1.33	1.36	1.26	1.39	1.80	2.17	2.27
	HDD10.0	872	146	147	148	113	67	16	1	1	4	28	80	122
	HDD18.3	3692	404	380	406	363	323	248	199	182	206	272	328	381
	CDD10.0	220	0	0	0	0	2	17	60	77	48	15	2	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.7	9.7	9.5	8.4	7.3	6.2	5.9	5.3	5.7	7.0	8.5	9.1	9.6
Precipitation	PrecAvg	997	119	92	84	53	48	52	56	77	82	110	114	112
	PrecMax	1222	216	165	152	81	114	105	119	210	139	158	189	193
	PrecMin	840	33	37	17	26	10	14	19	27	27	55	55	52
	PrecStd	113	45	33	30	14	21	23	26	36	27	30	34	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.3	9.6	9.7	10.5	13.5	14.3	17.4	17.3	15.2	13.2	11.9	10.7
		MCWB	8.5	8.7	8.3	8.8	11.8	12.4	16.0	15.4	13.7	12.2	10.9	9.9
	2%	DB	8.8	9.0	8.8	9.7	11.6	13.2	15.3	15.7	14.3	12.7	11.2	10.2
		MCWB	8.0	8.3	7.9	8.5	10.1	11.9	14.4	14.7	13.0	11.7	10.4	9.5
	5%	DB	8.3	8.3	8.3	9.1	10.7	12.5	14.5	15.0	13.8	12.2	10.7	9.7
		MCWB	7.5	7.5	7.4	7.9	9.5	11.4	13.7	13.9	12.7	11.3	9.9	9.0
	10%	DB	7.9	7.7	7.8	8.5	10.1	11.9	13.8	14.4	13.3	11.8	10.2	9.1
		MCWB	7.1	6.8	7.0	7.4	8.9	10.9	13.0	13.4	12.4	10.9	9.4	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	9.2	8.7	9.5	12.1	12.9	15.8	15.7	14.2	12.7	11.4	10.2
		MCDB	9.2	9.3	9.1	9.8	13.4	13.5	16.7	16.3	14.6	13.0	11.6	10.5
	2%	WB	8.2	8.5	8.2	8.7	10.6	12.1	14.6	14.9	13.5	12.1	10.7	9.7
		MCDB	8.5	8.8	8.6	9.2	11.2	12.7	15.0	15.7	13.9	12.4	11.0	10.0
	5%	WB	7.8	7.7	7.7	8.1	9.8	11.6	13.8	14.2	13.1	11.7	10.1	9.2
		MCDB	8.2	8.2	8.2	8.6	10.4	12.2	14.2	14.8	13.5	12.0	10.6	9.5
	10%	WB	7.2	6.9	7.2	7.6	9.2	11.2	13.2	13.7	12.7	11.2	9.6	8.5
		MCDB	7.8	7.6	7.7	8.2	9.8	11.7	13.6	14.1	13.1	11.7	10.1	9.1
Mean Daily Temperature Range	5% DB	MDBR	2.4	2.7	2.7	2.8	3.0	2.8	2.7	2.9	2.5	2.5	2.5	2.7
		MCDBR	2.6	2.8	2.8	3.0	3.5	3.2	3.5	3.4	2.7	2.5	2.6	2.8
		MCWBR	3.1	3.1	2.9	2.6	2.8	2.6	2.8	2.7	2.4	2.8	3.2	3.5
	5% WB	MCDBR	2.6	2.8	2.7	2.8	3.2	3.0	3.3	3.1	2.5	2.4	2.5	2.8
		MCWBR	3.1	3.2	3.0	2.5	2.7	2.6	2.8	2.7	2.4	3.0	3.2	3.6
Clear-Sky Solar Irradiance	taub	0.333	0.352	0.367	0.384	0.373	0.369	0.382	0.369	0.363	0.348	0.329	0.438	
	taud	2.112	2.228	2.296	2.303	2.406	2.455	2.430	2.468	2.449	2.339	2.168	2.536	
	Ebn at Noon	417	636	759	815	857	865	842	827	765	639	421	299	
	Edh at Noon	50	74	92	108	105	101	102	91	79	65	46	35	
All-Sky Solar Radiation	RadAvg	0.27	0.84	2.01	3.61	5.04	5.04	4.64	3.79	2.34	1.16	0.39	0.17	
	RadStd	0.03	0.10	0.14	0.28	0.33	0.45	0.40	0.33	0.18	0.11	0.03	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		+0.32	N/A	+0.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page